Swarm Mobile Android Performance

This document concludes the performance characteristics of the Swarm Mobile Android application during its development phase. The analysis is based on real phone usage and Android Studio telemetry captures, focusing on CPU and memory allocation. All measurements and observations are made with a Light Bee node.

Architecture

The application was implemented in Java for Android. It uses Golang's *gomobile* tool to compile the *bee-lite-java* library and be able to use the Bee code, which is implemented in Golang.
The app has two main components: 1. The UI Activity, 2. Background Service.

In short, the **Activity** starts the **Background Service** and interacts with it, and while the Activity gets into the background, the Service remains active and running on a different thread.

Note: to stop the background service, the whole task needs to be closed via the Android task manager

CPU Usage & Threading

Since the wrapped Bee node code uses peer-to-peer connections under the hood, it cannot be explicitly measured, but during download and upload these are the observations using Android Studio Profiler.

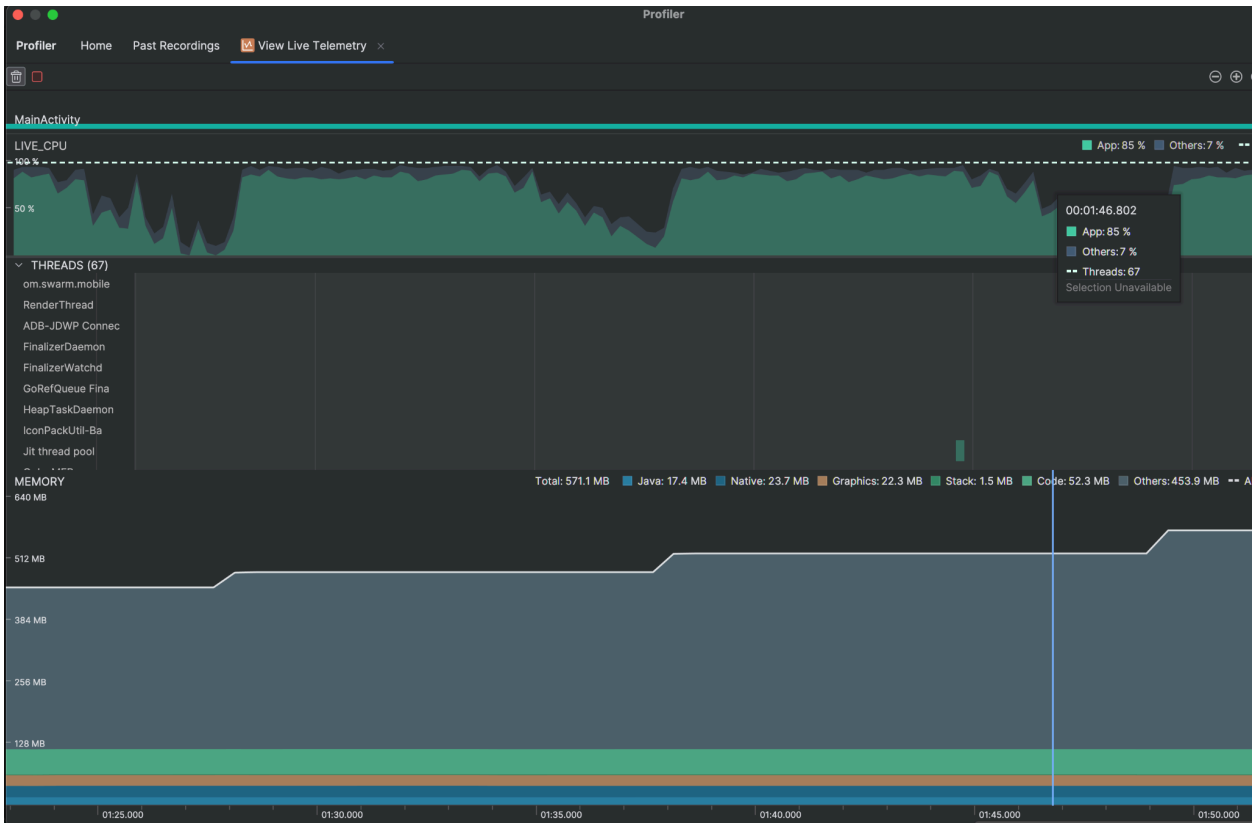
- **Utilization:** CPU usage frequently reaches peaks of **90%** during download (figure 1). While uploading the same file, it's using only around **50%** of CPU (figure 2).
- **Concurrency:** The app maintains between **32 and 72 active threads** simultaneously. These threads are controlled by:
    - `com.swarm.mobile`: Primary application thread.
    - Multiple generic `Thread-x` instances: Likely managing individual peer-to-peer node connections by the Bee node.

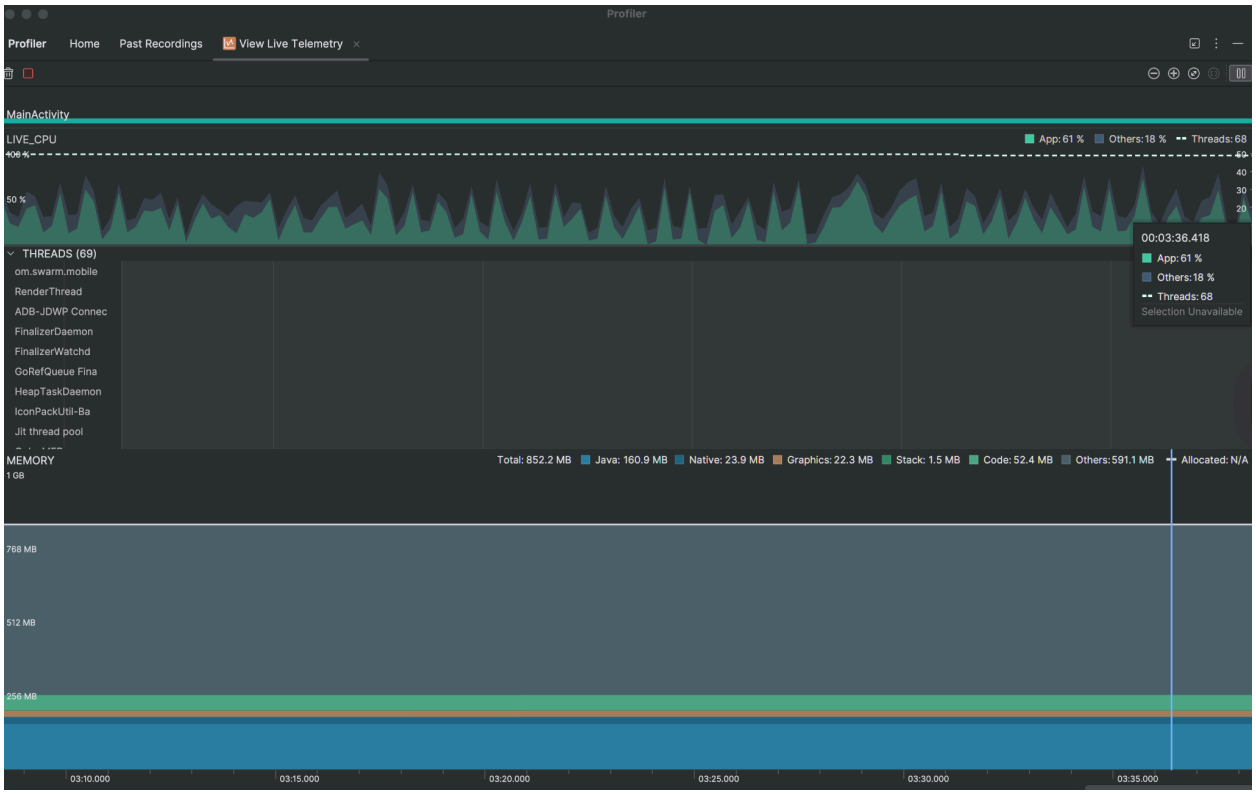
Memory Profile

The memory usage shows a "staircase" growth pattern, which is a key area for optimization.

- **Memory Footprint:** Typically ranges from 500 MB to 852 MB, and **never reaches 1GB**.
- **Allocation Breakdown:**
    - **Java Heap:** Relatively stable between 75 MB and 160 MB.

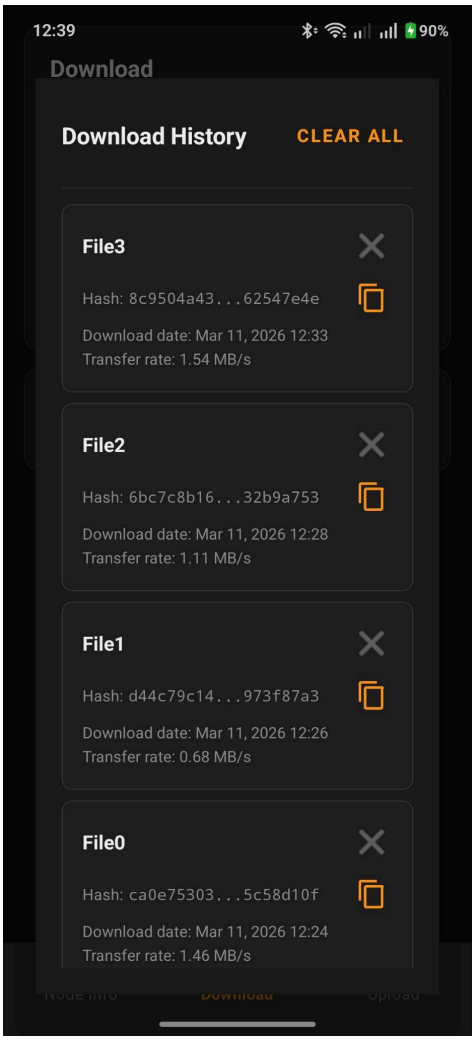
- **Others (Native/Buffers):** This is the largest segment, often exceeding **500 MB**.

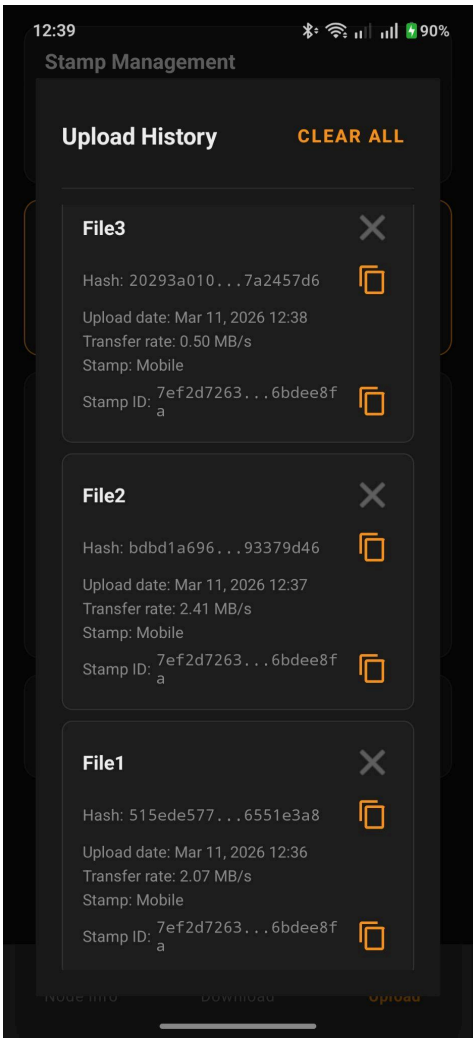




Network & Data Transfer

1. Over a network bandwidth of 1 Gbps with Wi-Fi 6:
    a. **Download Performance:** Peaks observed around **1.54 MB/s**.
    b. **Upload Performance:** Peaks observed as high as **2.41 MB/s**, often exceeding download rates - interestingly.





2. Over 5G mobile network:
    a. Both Download and Upload around **0.2MB/s-0.3MB/s**, and the same applies, as upload is a little bit faster than the download transfer rate.

Battery Usage

During a 3-hour-long continuous session of downloading and uploading data, the battery drain is about 10%.

Keeping the Light Bee node running as a background service, it drained 30% during an 8-hour time window, so it was not significant.